

Institute of Actuaries of India

Subject CP1 – Actuarial Practice (PaperA)

November 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

Financial Risk- if the business model is a failure and company is not able to recoup its invested capital or required return

Higher Expense Risk; if the expenses are higher than expected

Cyber risk; If the customer information is leaked then it would lead to breach of privacy, reputational and may leads to legal cost as well.

Inflation risk; IT cost increases at much faster pace than expected, or technology becomes obsolete over time leads to further expenses.

Competition; If the model is copied by someone else with aggressive strategy and deeper pockets then it may become difficult to sustain in business

IT risk – This is an application-based product that the Company is pitching to the prospective customers. Hence, there is a huge IT risk in terms to turnaround time, data safety, interface failure, connectivity etc.

Lower volume; Since the business is app based and hence its target will only be well educated and sophisticated customers who will be highly networked and any perception of poor quality/wrong advice/behaviour will lead to immediate adverse publicity/could go viral Inaccurate/Incomplete Information supplied by customer may mislead the services provided by the app and health coaches.

Operational fraud risk on part of employees and health coaches

Reputational risks – If customers are not satisfied with the services or some of the coaches advised inappropriately

Legal risk – wrong advice or improper exercise may lead to injuries which would lead to further legal issues if cases were filed against the company.

Regulatory risks – restriction on usage/collection of customer information.

Further, changes in tax laws also reduces company revenue.

[6 marks]

Solution 2:**Advantages**

High net worth individual his risk appetite would be high and hence having a portion of wealth earmarked for art should be fine

They should provide good diversification from other asset classes.

Return would mostly be from capital appreciation and there would be no regular income which ideally should suit an HNI investor

Likely to be more tax efficient as tax on capital gains would generally be lower than tax on income

Though there is no direct link of return to price inflation but value can be expected to increase in value and beat inflation if artist's popularity continues to grow

Feel good value from owning work of a popular artist

Disadvantages

Marketability could be poor as preferences/choices will differ vastly between are enthusiasts Prices of individual works of art are subject to volatility due to changes in taste/preferences. Valuation of a piece of art is pretty subjective

Expertise is needed to select and value works of art

Each piece of art is unique and actual value of piece of art is unknown unless sold Storage and security costs would be high reducing return

[6 Marks]

Solution 3:**i) Importance of ALM:**

Asset liability management ("ALM") is a fundamental element of retirement benefit scheme. A retirement benefit scheme works on the concept of taking regular contributions from employer/employee and then paying retirement benefits to a member on his attaining a retirement age

Hence, liabilities are long tailed in a retirement benefit scheme.

In order to ensure that the Scheme has sufficient assets to pay off the liabilities on time, it needs to invest in the assets that matches the liabilities by nature and duration.

Further, there are other liabilities such as regular expenses (like salary/consultant fee, member servicing expenses etc) that shall be considered and sufficient liquid assets shall be kept in the portfolio.

Due to the level of uncertainty of the existing liabilities (both in amount and timing) as there could be liabilities associated with death/early leavers/ill health retirement etc. there is a need for further liquidity.

[0.5 each, Max.3]

However, perfect matching is not always desirable and possible due to the following reasons

[2]

The sponsors' risk appetite

Level of funding of scheme

Desire to earn higher investment returns

Non-availability of asset classes to replicate the liabilities in nature and term
Statutory and regulatory restrictions

The need of diversification

[5]

ii) Consequences due to ALM mismatch:

Insufficient assets: If asset and liabilities are not matched, then there could be years when assets are not sufficient to pay off the liabilities. In this case, the Scheme could sell assets at loss and pay off the liabilities. Hence, value of assets gets eroded.

[1]

Illiquid assets: There could be a scenario at total level assets and liability are matching but assets are illiquid and can't be sold to pay off the liabilities. In which case there is risk of employee dissatisfaction, unrest and regulatory interventions.

[1]

Reinvestment risk: If asset duration is not matched with liability duration (e.g. asset duration is short term versus liability duration in long term) then there could be scenario when asset proceeds (maturity and coupons) are invested at lower yield compared to their current rate of return.

[1]

Interest rate risk: There is risk that future interest rates will be lower than current level or/and interest rate assumption used in determination of contribution rate. The interest rate at which the future contributions will be invested is unknown and could result in lower return than expected which lead to increase in contribution rates in future.

[1]

[4]

[9 Marks]

Solution 4:**i) It is a small company, hence some level of quota share agreement may be appropriate as-**

[2]

To increase its underwriting capacity

To reduce solvency capital requirement

to benefit from high initial reinsurance commission which would help in managing of expenses

- Excess of Loss may be required for Hull and Cargo separately for- [1]
 Protection of net account from aggregation
 of losses Large One off claims
 Stop Loss covers can also be considered to protect from significant upward movement in losses as
 being a small company high loss ratios is a particular cause for concern. [1]
 [4]
- ii) Financial reinsurance (FinRe) – a reinsurance arrangement that provides capital, typically by
 exploiting some form of regulatory, solvency or tax arbitrage. [0.5]
 It frequently relies on the regulatory, solvency or tax position of a reinsurer, which may be based
 in an overseas state, being different from that of the provider. [0.5]
 [1]
- iii) A typical example of exploiting a regulatory position is a contingent loan from the reinsurer to the
 insurance company. [0.5]
 A normal loan would increase the insurance company's assets, [0.5]
 But there would be no strengthening of the available capital position as the insurance company
 would also have to identify the amount owing as a liability. [1]
 With a contingent loan, the repayments are contingent on, for example, the insurance company
 making profits in future on a block of business. [1]
 Because the insurance company has no liability to repay the loan unless these profits emerge,
 the regulatory regime may allow it to not make any provision for these future payments on a
 statutory basis. [0.5]
 Therefore the insurance company improves its statutory solvency position. Bear in mind that, as
 the company will repay the loan over time, the improvement is only short term. [0.5]
 [4]
 [9 Marks]

Solution 5:

i)

The government could increase revenue from taxation or
 Distribute the revenue differently e.g. taking away funding from
 other areas They could use a general increase in taxes or
 have targeted taxes that apply for a given period (e.g. focusing on those who will
 benefit most from the programme)

The other method is that the government could borrow money.

Given that the project is likely to be medium to long term, the government will
 probably borrow by issuing bonds.

Although the country has capital markets, these may not have the required capacity for the issue It is
 not mentioned clearly how deep the capital market is and hence they may need to target
 overseas investors.

They may wish to arrange repayments to tie in with proceeds i.e. when the benefits of
 the programme lead to increased taxation revenue as a result of economic growth.

An option could be some form of partnership with the private sector.

In effect, the private sector would be responsible for funding and completing the project.

But they would receive fees from the government over a set period as compensation.

Such fees could be fixed (maybe inflation linked) or in the form of equity e.g. a share in toll
 revenues. [0.5 each, Max. 6]

ii) General risks

Some cities or areas may be missed out leading to local troubles. Noise, air pollution or environmental damage.

the project taking longer than expected to complete (delays)

Plans may involve destruction of buildings e.g. historical places, which could cause social unrest Bad weather or natural disasters.

Poor quality of work leading to high maintenance costs

Lack of use/lower use than expected leading to lower tolls than expected Default by contractors/their insolvency

If capital raised through issuance of bonds-

Risk of under subscription

Forced to offer high yields than anticipated increasing project costs overall

Tax increase/tolls means

Unhappy citizens leading to electoral unpopularity or protests

Inability to raise taxes/tolls further in case of cost overruns

[0.5 each, Max. 6]

[12 Marks]

Solution 6:

i) Principal aims of regulator of financial services

[0.5 each, Max 2]

to correct perceived market inefficiencies and to promote efficient and

orderly markets to protect consumers of financial products

to maintain confidence in the financial

system to help reduce financial crime.

ii) Meeting these aims

Market efficiency

[0.5 each, Max.1]

ensuring sufficient liquidity in the market place, eg by having market makers

the provision of settlement systems (to ensure that trades are carried out in an efficient and orderly way)

Consumer protection [Max.5] disclosure of information and product literature (to ensure that customers understand [0.5]

what they are buying e.g. unambiguous brochure and market literature, clear enunciation of

coverage and exclusions, waiting periods, warranties etc [1]

cooling off periods to retail customerse.g. 15 days period to the policyholder to cancel his policy if

he/she thinks that the policy does not cater his/her needs as retail policyholders should be given

time to go through the policy document and have an informed decision. [1]

GI customers are both retail and corporate. Corporates might not be as vulnerable to risks “mis-

selling” or “mis leading statements” like retail customers [0.5]

Control on intermediaries for eg., brokers, in terms of their responsibilities, delegation of underwriting and claims authority to them by company and how they should be exercised [0.5]

schemes to compensate investors for breaches of the regulations e.g. interest payments on delay

in settlement of claims [0.5]

legislation preventing the use of unfair contract terms by having approved policy contracts by the regulatory itself or by having standardized wordings and definitions especially for retail products

[0.5]

on premiums that are charges and reducing discriminatory practices [0.5]

Confidence [0.5 each, Max 1.5]

checking ongoing solvency (capital adequacy) of authorised providers e.g. to have a minimum solvency requirement in place

enforcing strict accounting information requirements using prescribed Accounting

policy ensuring the competence and integrity of financial practitioners

Financial crime reduction [0.5 each, Max 3.5]

vetting (investigating and approving) the firms authorised to conduct certain activities for eg., auditors, actuaries, brokers

vetting the individuals authorised to conduct certain activities for eg., key people like chairman of board, CEO, CFO, Appointed Actuary

Corporate governance standards

enforcing regulations

investigating suspected breaches of regulation

Imposing sanctions if regulations have not been met.

Inspections on regular intervals to check the market conduct

[10]

[12 Marks]

Solution 7:

- i) A claim is payable on this contract when the policyholder is deemed to have reached a specified level of disability, for example the policyholder may be unable to perform a specified number of “activities of daily living” (ADLs). [1]

The different levels of care will differ between one contract and another, but typically may include: [2]

- cost of care in own home
- cost of being cared for in a residential (but non-nursing) home
- cost of being cared for in a residential nursing home.

The contract can be paid for by single or regular premiums, [0.5]

And can be offered under any platform-without-profit, with-profit and unit linked [0.5]

[4]

ii)

Benefit Design / Structure [2]

- o Understand Customer need :
- o Fixed vs Increasing benefit product to keep pace with inflation
- o Which platform- non-participating/participating or unit linked with guarantees
- o Target Market , customer sophistication

Pricing [2]

- o Availability of Data
- o Assumptions: Demography e.g. -longevity rate/ Economic, Trends / Profitability
- o Inflation-expense inflation, medical inflation rate
- o Option and Guarantees- surrender, paid up,

Shareholder Risk appetite	[1]
o Capital Requirement, available capital, is this the optimal use of capital	
o Profitability target considering tax savings offered by Government	
Reinsurance support available	[1]
o For pricing	
o Initial/claims underwriting	
Competition	[1]
o Key competitors and likely strategy	
o Price sensitivity of market	
o Risk of adverse publicity from uncompetitive rates	
Regulation and Legislation	[1]
o Likely to be pressure from the Government to come up with the competitive rates	
o Limitation and restrictions imposed by Regulations (May be relaxed since it's a new kind of product and Government driven initiative)	
o Compliance	
o State Benefits if any and offset required	
Marketability	[1]
o Government is actively promoting these products which would help	
o Tax incentives to insured needs to be publicised	
o Branding	
o Strategy	
o Distribution Channel e.g. online may purchase can have lower or nil commission	
Administration Pre / Post Event	[1]
o Need to build network of medical professions for claims assessment and management	
o To decide whether Services- outsourcing/internally managed	
System development - New product and hence time and costs need to be considered	[0.5]
	[Max.9]
	[13 Marks]

Solution 8:

- i) Mortality experience study is done to compare the actual death claims versus expected death claims in a given period.
First of all, select the period for which mortality investigation need to be done. E.g for last 5 years, last 2 years etc.
The data shall be selected in a manner that there are no significant changes in underwriting practices, product types, target market and claims management processes.
While selecting the data, identify any random fluctuations, abnormal fluctuations and data errors. If possible, avoid/adjust them.
- Next step is to splitting the data into homogeneous groups e.g: [1½]
- 📊 Male /Female
 - 📊 Age band
 - 📊 Underwritten/Non-underwritten policies
 - 📊 Channel wise – Direct sales and Bank partnerships
 - 📊 Sum assured band/Average Ticket Size wise etc
 - 📊 Product/LoB wise
 - 📊 Duration

 **Smoker/Non smoker**

 **Occupation**

While sub-dividing the data, ensure that there is sufficient and credible data in each classification. There may have been changes in target market (socio-economic changes) and hence the data should be carefully used.

For each classification, calculated death outgo and compare it with actual death outgo.

Ideally, mortality experience is monitored policy year wise (duration for entry). Example: Claims expected in Policy year 1 versus actual claims within Policy year 1 and so on.

By doing analysis by policy year wise will identify any trend of early claims and hence, corrective measure can be taken.

[0.5 each, unless stated otherwise, Max.6]

ii) Factors:

A term insurance product aims to provide only coverage in case of death of life insured during the term of the policy.

Generally, a term insurance product is sold for a high sum assured. Hence, underwriting plays an important role to select good lives for the coverage.

Hence, the target market for a term insurance product is usually different compared to endowment assurance product.

Factors of variations are – Income level, health conditions, age, sex, marital status etc.

On other side, the aim of an endowment assurance is saving cum protection. A major part of the premium is saving element and a smaller proportion is protection element.

Underwriting requirements are generally less in an endowment plan compared to term plan.

Expect overall lower term assurance mortality than endowment because of stricter underwriting in term plans.

However, there can be factors revering the fact for example affordability of endowment products, target market, minimum premium size etc.

[0.5 for each relevant point,Max.4]

iii) Rating Factors:

Gender: Usually, lower mortality risk is expected for females.

Age: Lower mortality risk is expected for younger people.

Lifestyle: This includes if the customer is active regularly and eating healthy.

Hobbies: Hazardous hobbies such as mountaineering indicates more mortality risk
Smoker/Non-smoker:

Drinking habits

Occupation and Income level

Medical History/Family history:

Location:

[0.5 each,Max. 4]

[14 Marks]

Solution 9:

i)

The claims outgo may be volatile due to:

- Volatility in incidence rates or
- Volatility in claim amounts or
- Both

[1]

Volatility in Incidence rates may be due to:**[3]**

Change in Composition of group - i.e. age / gender mix, more married people
The average age of the portfolio may be increasing which could increase the volatility in incidence rates and

Morbidity increases with age and sex

Changes in economic cycle may have forced the employer to restructure business and hence the composition of group may have changed as a result.
Impact of the wellness or preventive care initiatives - e.g. free cancer screening camp. Such initiative could result into a sudden increase in claims
Accidents / pandemic / catastrophe events - The concentration risk is very high within a group. Any catastrophe or pandemic can severely impact the overall claims experience.
Incidence rates are increasing in general due to medical advancement of screenings, or it may be due to external factors such as high pollution levels or global warming

Volatility in Claim amounts may be due to:**[3]**

Medical inflation - higher than anticipated cost of medicals due to better facilities or in general cost would have increased.

Medical and technological advancements - newer diseases or treatments may lead to unexpected claim cost.

Provider behaviour - the employer has very little control over the provider services and billing behaviour and hence any change in treatment or billing protocol would impact the claim cost.

Poor Claims handling and management may also lead to higher claims

Large claims - since the benefits are unlimited, periodic large claims would add to the volatility. Unnecessary treatments or frauds

[7]**ii) Mitigation**

The Company may take insurance of some or all of the benefits from a insurance provider. The company pays a fixed premium in a given year to transfer the volatility risk to an insurer

Change in benefit structure: The company can re-design the benefit structure by introducing benefit ceiling.

This may be made dependent on the type of illness and to reduce unnecessary treatment and frauds as well.

To reduce the claim cost, the company may introduce coinsurance whereby health scheme cost will be shared with the employee

Currently, the Company cover both inpatient and out-patient cost. The Company may plan to reduce the coverage by limiting itself to inpatient coverage only.

The Company may plan to restrict on dependent coverage

Change in eligibility criteria- the employer can introduce a stricter minimum eligibility criterion Stringent claims management will ensure that only the eligible claims are paid.

More focus on wellness and preventive care is likely to reduce large claims if the condition / diseases are diagnosed at an early stage.

[0.5 for each relevant point,Max.5]**iii) Insurance product to buy:**

First of all, it is important to decide what type of benefits the Company would like to provide to its employees.

Does the coverage should include inpatient and / or outpatient treatment and / or medicine cost? coverage should be provided to employees' spouse and kids?

coverage should be provided to employees' parents?

Is Company ready to bear the cost for this or employee will bear this cost?

The next step will be to decide the structure of the insurance cover.

A group health insurance policy may be ideal to buy for the market which have the similar cover as the existing scheme.

.....But generally, such policies come with limited sum assured.

The Company may plan to introduce maximum sum insured for health insurance scheme. The maximum limit may vary depending on employees' designation/level.

There may be other requirements (waiting period, pre-existing conditions etc.) imposed by the insurer.

Further, the Company may decide to sponsor full premium or partially paid by the employees.

The other option with the Company to limit liability and reduce volatility by buying Individual Excess of loss (buying top ups with a deductible), on an aggregation of claims (Group Top Up) or on a catastrophe basis.

Purchase of such covers limits an employer's liability to a set Rupee figure per individual, per policy year.

The company can also choose a combination of individual excess of loss cover and stop loss cover to limit the overall liability.

[0.5 each for each point, Max.7]

[19 Marks]
